Word Codes (Set B)

Year 4 Pt 2 · **Mark Scheme**

Lesson Worksheet — Answers

Spotting the Code

1. COG — shifting each letter BACK by 1: D→C, P→O, H→G.
2. DBU — shifting each letter forward 1: C→D, A→B, T→U.
3. A (QJH) — shifting forward 1: P→Q, I→J, G→H.
4. A, B, C, D, E — every pair listed is correctly shifted forward by 3.
5. DAT — shifting each letter BACK 1: E→D, B→A, U→T.
6. A (SVO) — shifting forward 1: R→S, U→V, N→O.
7. JGP — shifting forward 2: H→J, E→G, N→P.
8. IOG — shifting each letter BACK 2: K→I, Q→O, I→G.
9. IAT — shifting each letter BACK 1: J→I, B→A, U→T.
10. A, B, C, D — each correctly reverses the +1 shift by moving back 1; E is incorrect (K→J, Q→P, I→H gives JPH, not LPH).
11. QJH — shifting forward 1: P→Q, I→J, G→H.
12. A (AZS) — shifting back 1: B→A, A→Z (wrapping), T→S.
13. JEX — shifting each letter BACK 2: L→J, G→E, Z→X.
14. FDW — shifting forward 3: C→F, A→D, T→W.
15. A, B, C, D — each is correctly shifted forward by 2; E is incorrect (C→E, A→C, T→V, giving ECV, not EDV).

Applying the Code

16. The shift is +2 (C→E, A→C, T→V). Encoding DOG with +2: D→F, O→Q, G→I, giving the code 'FQI'.
17. CON — shifting back 4: G→C, S→O, R→N.
18. SWIM — shifting forward 3 (undoing a back-3 shift): P→S, T→W, F→I, J→M.
19. A, B, C, D, E — every pair is correctly shifted forward by 5.
20. The shift is +2 (B→D, I→K, R→T, D→F). Encoding NEST with +2: N→P, E→G, S→U, T→V, giving 'PGUV'.
21. TIN — decoding PEJ by shifting forward 4 (undoing a back-4 shift): P→T, E→I, J→N.
22. WIND — decoding ZLQG by shifting BACK 3: Z→W, L→I, Q→N, G→D.
23. BTQK — encoding WOLF with a +5 shift: W→B (wrapping), O→T, L→Q, F→K.
24. A, B, C, D, E — every pair is correctly shifted back by 6 (wrapping where needed).
25. The shift is +2 (D→F, U→W, C→E, K→M). Decoding JQWUG by shifting BACK 2: J→H, Q→O, W→U, U→S, G→E, giving 'HOUSE'.
26. LOYN — encoding FISH with a +6 shift: F→L, I→O, S→Y, H→N.
27. FISH — decoding ILVK by shifting BACK 3: I→F, L→I, V→S, K→H.
28. BIRD — decoding UBKW by shifting FORWARD 7 (undoing a back-7 shift): U→B, B→I, K→R, W→D.
29. A, B, C, D, E — every pair is correctly shifted forward by 4.
30. The shift is +3 (H→K, O→R, R→U, S→V, E→H). Encoding SNAKE with +3: S→V, N→Q, A→D, K→N, E→H, giving 'VQDNH'.

Codes in Context

31. SHARK — the shift is +3 (T→W, I→L, G→J, E→H, R→U). Decoding VKDUN by shifting BACK 3: V→S, K→H, D→A, U→R, N→K, giving 'SHARK'.
32. FBD — C is the 3rd letter, doubled = 6 = F; A is the 1st letter, doubled = 2 = B; B is the 2nd letter, doubled = 4 = D.

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33. A, B, C, E — shifting CAT forward 2 gives C→E, A→C, T→V, i.e. 'ECV' (confirming A); reversing 'ECV' gives 'VCE' (confirming B), which is also the final coded word (confirming C, so D is wrong); reversing CAT first gives 'TAC' (confirming E).

34. CBT — the vowel A shifts forward 1 to B; the consonants C and T stay the same.

35. RIVER — the shift is +3 (L→O, E→H, M→P, O→R, N→Q). Decoding ULYHU by shifting BACK 3: U→R, L→I, Y→V, H→E, U→R, giving 'RIVER'.

36. XZG — reversing the alphabet: C(3rd) ↔ X(24th, since 27−3=24), A(1st) ↔ Z(26th), T(20th) ↔ G(7th, since 27−20=7). So 'CAT' becomes 'XZG'.

37. A, B, C, D, E — every mapping is correct: L→J, E stays E, M→K, O stays O, N→L (each consonant shifted back 2; vowels unchanged).

38. 13, 11, 12 — C is the 3rd letter (+10=13), A is the 1st letter (+10=11), B is the 2nd letter (+10=12).

39. The shift is +3 (G→J, R→U, A→D, P→S, E→H). Encoding LEMON with +3: L→O, E→H, M→P, O→R, N→Q, giving 'OHPRQ'.

40. CMSQF — shifting HOUSE back 2 gives 'FMSQC'; swapping the first and last letters (F and C) gives 'CMSQF'.

41. A, B, C, D — reversing the alphabet: B(2nd) ↔ Y(25th), A(1st) ↔ Z(26th), T(20th) ↔ G(7th), giving the coded word 'YZG'; E reverses the letter order incorrectly, so it is wrong.

42. E — E is the 5th letter; 5×3=15, and the ones digit of 15 is 5, which corresponds to the letter E (so E stays E under this rule).

43. The shift is +3 (A→D, P→S, P→S, L→O, E→H). Decoding VSULQJ by shifting BACK 3: V→S, S→P, U→R, L→I, Q→N, J→G, giving 'SPRING'.

44. MUASI — vowels move to the next vowel in the cycle (O→U, U→A, E→I); consonants M and S stay the same.

45. UBD — reversing CAT gives TAC; shifting each letter of TAC forward 1: T→U, A→B, C→D, giving 'UBD'.

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Homework Worksheet — Answers

Quick Practice

1. EAT — shifting each letter BACK by 1: F→E, B→A, U→T.
2. EPH — shifting forward 1: D→E, O→P, G→H.
3. A (SVH) — shifting forward 1: R→S, U→V, G→H.
4. A, B, C, D, E — every pair listed is correctly shifted forward by 2.
5. PIG — shifting each letter BACK 1: Q→P, J→I, H→G.
6. A (DBU) — shifting forward 1: C→D, A→B, T→U.
7. HQZ — shifting forward 2: F→H, O→Q, X→Z.
8. HEN — shifting each letter BACK 2: J→H, G→E, P→N.
9. RUN — shifting each letter BACK 1: S→R, V→U, O→N.
10. A, B, C, D — each correctly reverses the +2 shift by moving back 2; E is incorrect (E→C, D→B, V→T, giving CBT, not CAT).
11. IFO — shifting forward 1: H→I, E→F, N→O.
12. A (CNF) — shifting back 1: D→C, O→N, G→F.
13. COG — shifting each letter BACK 3: F→C, R→O, J→G.
14. RKI — shifting forward 2: P→R, I→K, G→I.
15. A, B, C, D — each pair is correctly shifted forward by 1; E is incorrect (B→C, A→B, T→U, giving CBU, not CAU).

Mixed Practice

16. The shift is +2 (D→F, O→Q, G→I). Encoding CAT with +2: C→E, A→C, T→V, giving 'ECV'.
17. WAT — shifting ZDW back 3: Z→W, D→A, W→T.
18. TIIR — shifting VKKT back 2: V→T, K→I, K→I, T→R.
19. A, B, C, D, E — every pair listed is correctly shifted forward by 4.
20. The shift is +2 (F→H, I→K, S→U, H→J). Encoding CRAB with +2: C→E, R→T, A→C, B→D, giving 'ETCD'.
21. OLUE — shifting ROXH back 3: R→O, O→L, X→U, H→E.
22. STAR — decoding VWDU by shifting BACK 3: V→S, W→T, D→A, U→R.
23. DIFVE — encoding ZEBRA with +4: Z→D (wrapping), E→I, B→F, R→V, A→E, giving 'DIFVE'.
24. A, B, C, D, E — every pair is correctly shifted back by 5 (wrapping where needed).
25. OHMER — the shift is +2 (C→E, R→U, A→C, B→D). Decoding QJOGT by shifting BACK 2: Q→O, J→H, O→M, G→E, T→R.
26. LTFY — encoding GOAT with +5: G→L, O→T, A→F, T→Y.
27. SNAKE — decoding VQDNH by shifting BACK 3: V→S, Q→N, D→A, N→K, H→E.
28. FQCRD — decoding ZKWLX by shifting FORWARD 6 (undoing a back-6 shift): Z→F, K→Q, W→C, L→R, X→D.
29. A, B, C, E — checking each with +3: BEAR→EHDU (matches A); SWAN→VZDQ (matches B); GOAT→JRDW (matches C); LION+3 gives OLRQ, not the ORLQ stated in option D, so D is wrong; DEER→GHHU (matches E).
30. The shift is +3 (M→P, O→R, U→X, S→V, E→H). Encoding TIGER with +3: T→W, I→L, G→J, E→H, R→U, giving 'WLJHU'.

Standard / Advanced

31. SNAKE — the shift is +3 (L→O, E→H, M→P, O→R, N→Q). Decoding VQDNH by shifting BACK 3: V→S, Q→N, D→A, N→K, H→E.
32. EAC — C is the 3rd letter: 3×2−1=5=E; A is the 1st letter: 1×2−1=1=A; B is the 2nd letter: 2×2−1=3=C.
33. A, B, C, E — shifting CAT forward 2 gives 'ECV' (confirming A); reversing 'ECV' gives 'VCE' (confirming B), which is also the final coded word (confirming C, so D is wrong); reversing CAT first gives 'TAC' (confirming E).
34. DNG — the vowel O shifts back 1 to N; the consonants D and G stay the same.

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35. FIVER — the shift is +3 (R→U, I→L, V→Y, E→H, R→U). Decoding ILYHU by shifting BACK 3: I→F, L→I, Y→V, H→E, U→R, giving 'FIVER' (a five-pound note).

36. WLT — reversing the alphabet: D(4th) ↔ W(23rd, since 27−4=23), O(15th) ↔ L(12th, since 27−15=12), G(7th) ↔ T(20th, since 27−7=20). So 'DOG' becomes 'WLT'.

37. A, B, C, D, E — every mapping is correct: T→U, I stays I, G→H, E stays E, R→S (each consonant shifted forward 1; vowels unchanged).

38. 8, 30, 14 — D is the 4th letter (×2=8), O is the 15th letter (×2=30), G is the 7th letter (×2=14).

39. The shift is +3 (S→V, N→Q, A→D, K→N, E→H). Encoding TIGER with +3: T→W, I→L, G→J, E→H, R→U, giving 'WLJHU'.

40. FPVTN — shifting MOUSE forward 1 gives 'NPVTF'; swapping the first and last letters (N and F) gives 'FPVTN'.

41. A, B, C, D — reversing the alphabet: D(4th) ↔ W(23rd), O(15th) ↔ L(12th), G(7th) ↔ T(20th), giving the coded word 'WLT'; E gives the letters in the wrong order, so it is wrong.

42. H — N is the 14th letter; 14×2=28, and the ones digit of 28 is 8, which corresponds to the letter H.

43. BEARS — the shift is +3 (T→W, R→U, A→D, I→L, N→Q). Decoding EHDUV by shifting BACK 3: E→B, H→E, D→A, U→R, V→S.

44. TEGAR — vowels move to the previous vowel in the cycle (I→E, E→A); consonants T, G and R stay the same.

45. FNC — reversing DOG gives 'GOD'; shifting each letter of GOD back 1: G→F, O→N, D→C, giving 'FNC'.

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